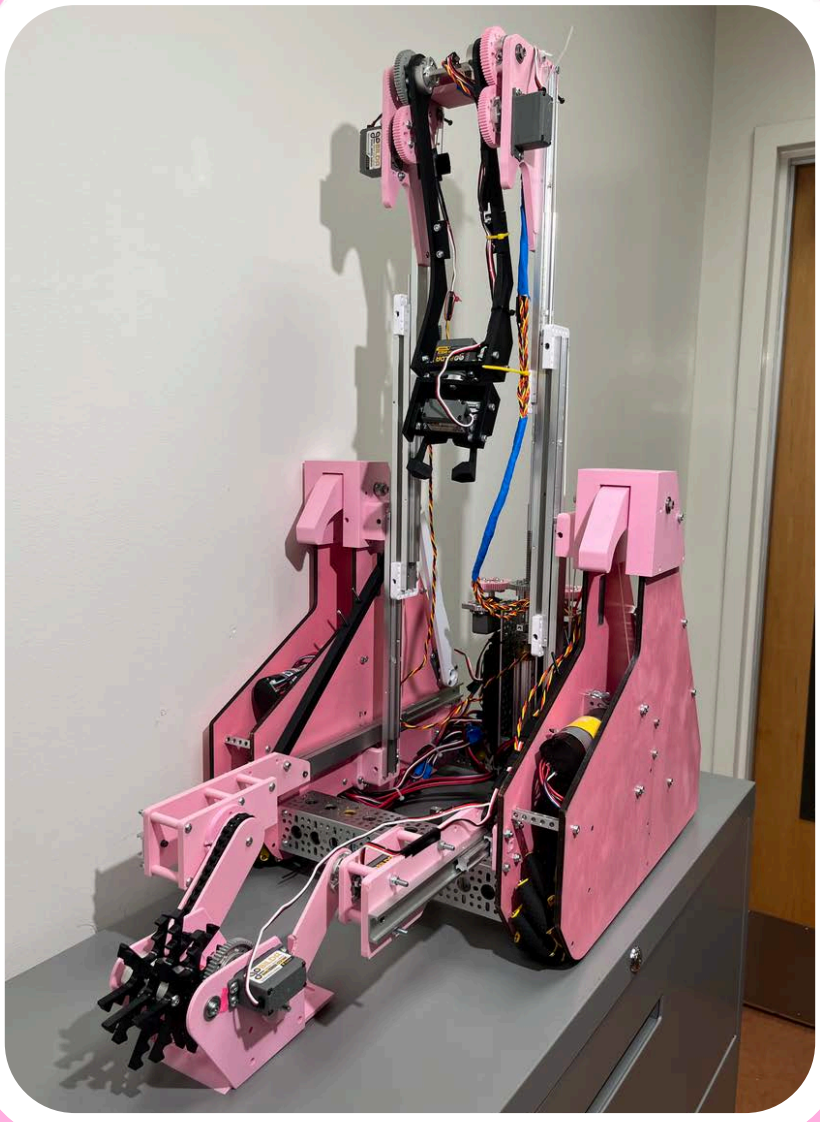
Team 51

Nerd Herd:
INSOMNIACS



Introduction

Our Team:

Team 51 CAMS Nerd Herd Insomniacs is a FIRST Tech Challenge (FTC) team based in Carson, California.



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Our Goal:

At our school, Nerd Herd strives to provide students with hands-on, real-world experiences in engineering and business. Outside of the classroom, Team 51 aims to spread STEM values throughout the community. We hope to inspire the next generation of problem solvers, go-getters and creative thinkers. Specifically within our team, we strive to create an environment that emphasizes teamwork, respect, innovation, commitment and gracious professionalism.

Our team focuses on developing the technical and personal skills of our members. We focus on design principles and processes, moving from concept to finished product. In our community, we consistently demonstrate our core FIRST values and gracious professionalism when working with others. We strive to increase the impact of FIRST and STEM through awareness. Additionally, we strengthen connections with technical experts and other members of the STEM community through collaboration.

Meet The Team

FTC Team 51 Insomniacs is composed of 9 students, 3 parent mentors, and 2 team leads. The whole team branches out to perfect each of our 4 subteams: Programming, Design, Assembly, Admin



Felix Li

Team Lead



Insomniac

Robot!



Sarah Arata

Team Lead



Aadi Jain

Programming



Madeleine Jiang

Design
Programming



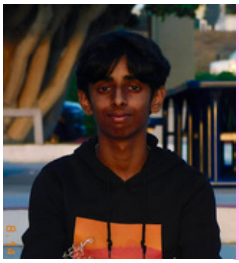
Michelle Adewinmbi

Design



Ashley Wills

Admin
Assembly
Design



Suyash Karajagi

Assembly
Admin



Micka Agacite

Assembly



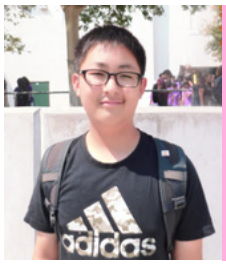
Emiliano Cruz

Admin



Sachi Yagi

Assembly



Brendan Lim

Assembly



Jaspal Sahdev

Parent Mentor



Robert Wills

Parent Mentor



Nick Jiang

Parent Mentor

Team Goals

The goals that we planned to achieve going into this year's FTC season were to:

- Gain more experience within the field of STEM
- Build communication skills across the team
- Have our team members collaboratively solve problems by approaching challenges with a strong, positive mindset
- Embrace sustainability with a mainly freshman team

Women in STEM

This year, we have furthered female representation within STEM by having an all-female CAD team. Throughout our experience this year, we have been building on each others' skillsets and encouraging each other to grow, which inspired us to continue CADing and designing intricate and innovative designs throughout the season.











Thank you to our Sponsors!

Raising Cane's Chicken Fingers

HONDA

Raytheon Technologies

BOEING

The Arata Family

Community Outreach

Our Theme This Season: **Sustainability**



Rummage Sale Volunteering

- We volunteered at a rummage sale in order to make our community more sustainable by promoting recycling!
- At this event, we helped sell second-hand goods at high discounts
- **The proceeds (~$1000) were donated to a library**



Recycled Material Plates

- Our realization: Cardboard was abundant all around us and we needed wood for our side-plates
- Our solution: Turn the cardboard into our number plates
- To do this, we used a combination of vinegar, milk, and cardboard in order to make solid boards
- We **shared this with the community through a tutorial** on how to make the boards

Plastic Bottle Recycling

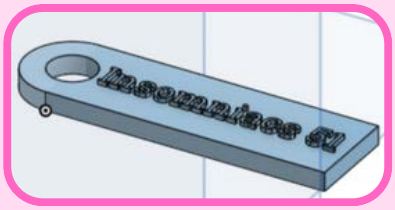
- As a part of our team's theme of sustainability, we decided to turn plastic bottles into the one thing our robot uses the most: 3D printer filament
- We took scrap parts from old and out of use 3d printers and modified its parts.
- Thanks to our plastic bottle recycler, we were able to **reuse around 25g of plastic**

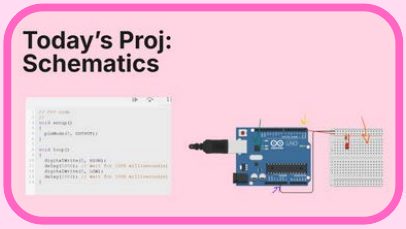




Online Lessons

- In order to make FTC more sustainable, we held online lessons to reach students who were interested in FTC-related skills and cater to their interests in CAD and programming.
- We taught CAD lessons using Onshape and programming lessons on Tinkercad.
- Through these lessons, we were able to reach a wider audience of students who had no prior experience and teach them how to create a basic keychain to further their interests in FTC and STEM





Community Events

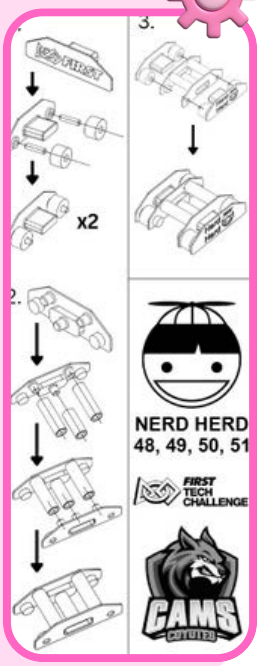
Impact on Younger Generation

- We gave a **presentation about FTC and STEM** to ~50 students at Point Vicente Elementary School, where we explained the FTC program, our team structure, the design process, and essential technical skills that we gain from FTC.
- At this presentation, we **passed out small robot kits to each student** with instructions. We designed, manufactured, and packaged these kits ourselves.
- The robot kits gave the kids fun, hands-on experience with STEM that they could take home and show their parents.













CAMS Site Day

- At CAMS Site Day, our team did robot demos, explained the FIRST program, and shared our experiences with 20+ future generation CAMS students.
- During our outreach, we spoke to the students about what we do individually and as a team in FTC. We showed them our CAD of custom parts and then the real-life counterparts to show what they could do in FTC, and then advertised our STEM classes to further their interests.

Beach Blitz

- We shadowed our school's FRC Team (go 687!) during their offseason match at Beach Blitz. We also ran an FTC scrimmage with our sister teams.
- Alongside running the scrimmage, we also did outreach with the FLL students who were there for a competition! They got to drive our robot on the field and score field elements!
- At this competition, we passed out around 10 mini-robot kits to the FLL teams





Industry Professionals

As we worked towards our goal of a successful robot, we also looked to connect with industry professionals in the community. We were able to get in touch with two engineers, and we received important advice and feedback that allowed us to improve our design even further.



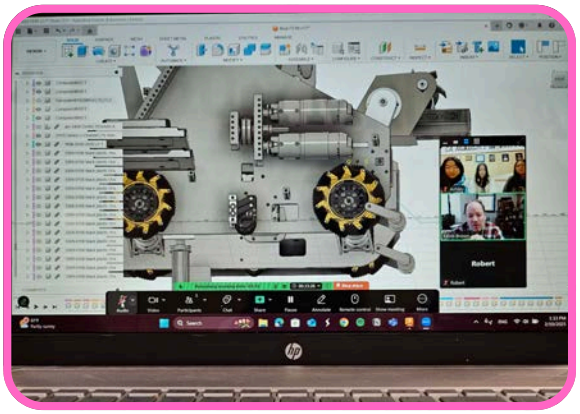
Dr. Andrew Truxel

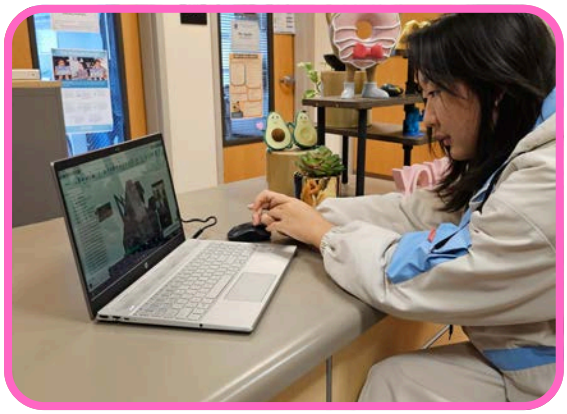
- Dr. Truxel is a **mechanical engineer at RTX**, as well as a **former mentor of FRC Team 4201 Vitruvian Bots**.
- He was able to help us **solve our center-of-mass issues** in our robot.
  - Our problem: We didn't know how to find the center of mass so we could balance the robot under our hooks
  - His solution: He helped us accurately **find the center of mass on our robot**.
  - While we experienced difficulties in shifting components of our design to center it, we were able to make use of our intake to change and balance our robot for a proper Level 3 Ascent thanks to his advice.

Dr. Kevin Brown

- Dr. Brown is an MIT Graduate with a **Doctorate in mechanical engineering**.
- He runs his own company named **Brown Innovations** and does directed and focused audio.
- We had a Zoom meeting with him where we discussed the **pros and cons** of our current robot design.
- He helped us with **planning and organizing** how we wanted to tackle each problem step-by-step by creating a Google spreadsheet to **maximize scoring efficiency.**



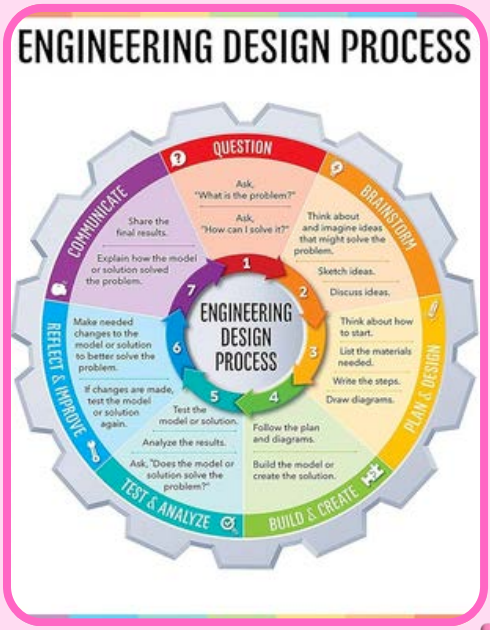




Design Principles

Here is the process we used to improve upon our robot:

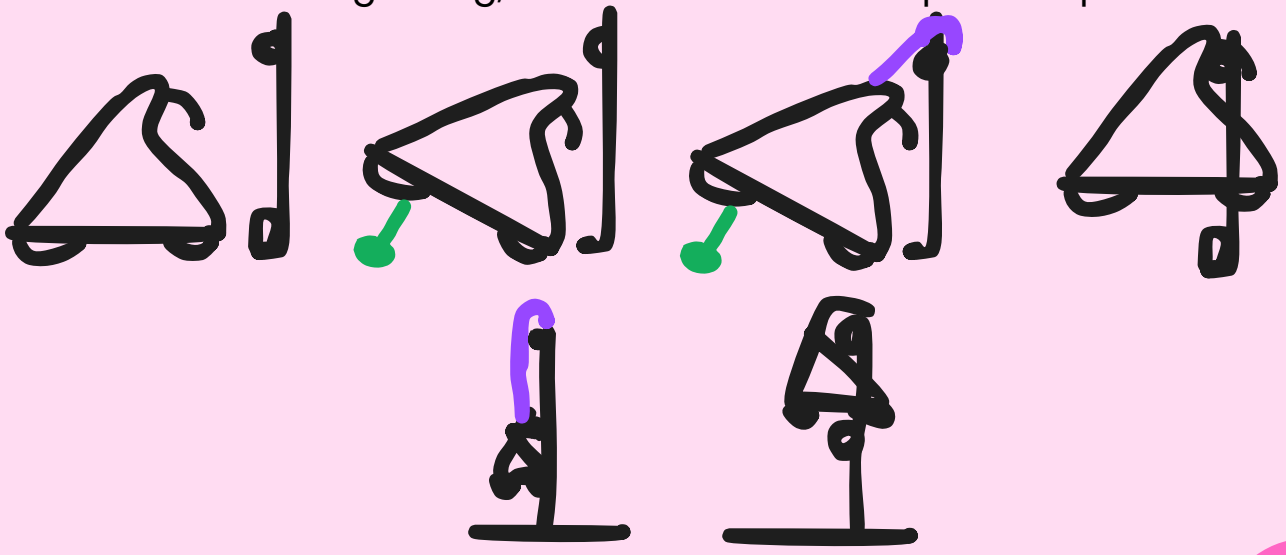
Question: What can we do to be able to perform a Level 3 Ascent?
Brainstorm: We could use a combination of top hooks and spring hooks to climb the rungs one at a time.
Plan and Design: We can use Fusion 360 to model and test the hooks virtually.
Build and Create: We used 3D printers to 3D print our designs and add the additional hardware.
Test and Analyze: We can add the hooks to our robot and test a Level 3 Ascent.
Reflect and Improve: We found that our robot struggled to center its mass properly to align the hooks to the rungs. The spring hooks also couldn't deploy properly.
Communicate: We talked about the issues and coordinated solutions for each problem with the team.
Question: How can we properly balance our robots weight when trying to hang on a rung?



Unique Features

Our robot consists of many creative subsystems that work together to help us play this year's game, but out of all systems there is one distinctive feature that we our most proud of: **Our Level 3 Ascent.**

To ascend to the high rung, we needed to come up with a process:



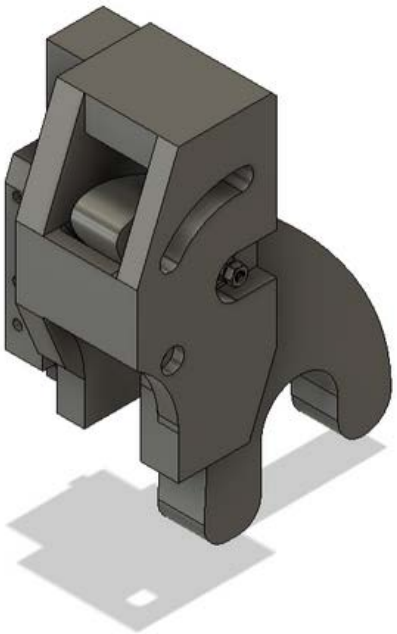
Iterative Spring Hook

Iteration 1

The spring hooks are the most unique part of our robot. Its simple mechanism but intricative design is what stands us out from the other competitors. It uses a hook with an attached spring at the corner to pushback, flatten, and slide past the 2nd level ascent bar, hooking on and allowing our top rung hooks to rise to the 3rd ascent. This is our first iteration, where we only had a rough idea of what we wanted the hooks to be.



Too big and chunky!



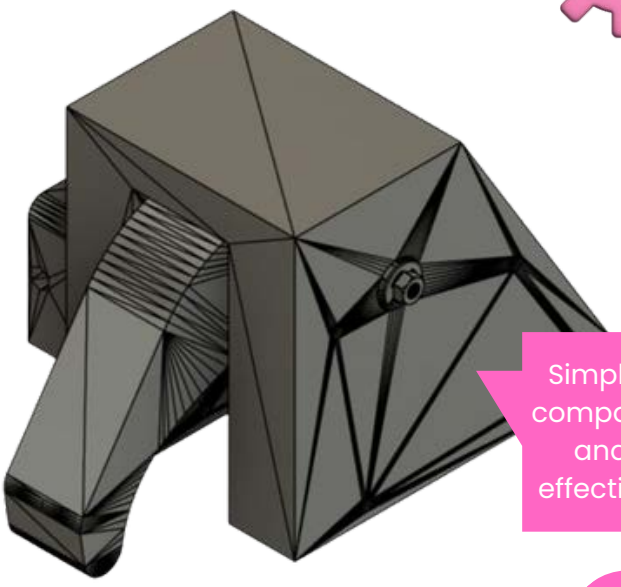
Not quite right for our new robot..

Iteration 2

For our second iteration of the spring hook, we added more stability to the hooks casing by connecting it at the top, and also made it a little less bulkier by compacting the casing in more. The hooks remained the same, but this design was much sturdier, saved space on our robot, and was also easier to assemble. However, the hook wasn't able to spring back out once flattened, as it hit the bar.

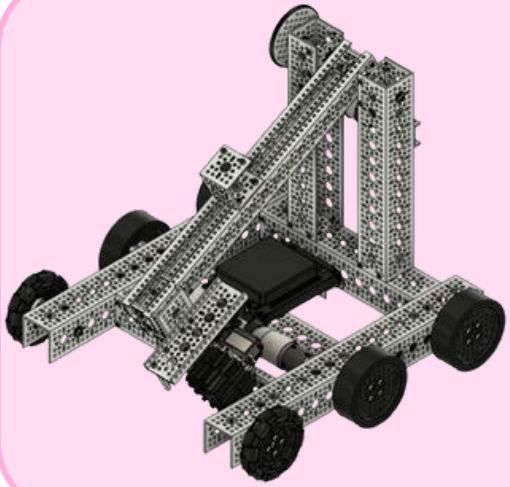
Iteration 3

Eventually, following our redesign, we settled on our final spring hook design. The smaller casing saves on space and height, and the smaller hook ensures that it doesn't get caught on the lower bar. It also integrates well into the new design, as it supports the outtake linear slides while also acting as a support between our inner and outer sideplates.



Simple, compact, and effective!

Robot Iteration



Pros:
- Simple and easy to build design
- Good for drive practice/assembly experience
- Gave more time for Design to create better designs

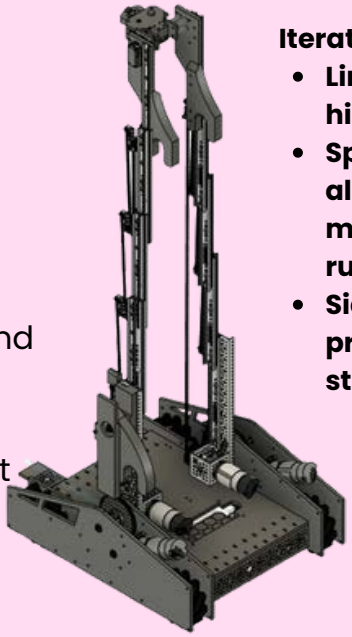
Cons:
- Not efficient for cycling
- Not driver friendly
- Can't reach high basket

Version 1

Unique Aspects
- Roller intake w/ wrist for specimens and samples
- Level 1 ascent with arm

Pros:
- Claw intake and linear slides allow for high basket scoring
- Hooks allowed for Level 2 Ascent

Cons:
- Claw was hard to aim and intake quickly
- Center of mass issues prevented Level 3 Ascent
- Weight of the slides put strain on the motors



Iterations from Version 1:
- **Linear slides allowing higher reach**
- **Spring hooks allowing the robot to manuever rung to rung**
- **Side plates preventying getting stuck on specimens**

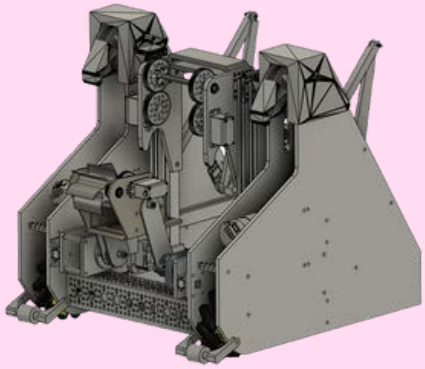
Version 2

Unique Aspects
- Rotating linear slides serve as intake/ outtake
- Spring hooks create Ascent opportunities
- Claw intake

Pros:
- Eco-friendly components
- Lighter/stronger linear slides
- Easier intake to use
- More reliable Ascent
- Sturdier frame

Cons:
- Intake can only pick up parallel samples
- Intake/outtake handoff is complicated

Iterations from Version 2:
- **more compact design helping better ascent**
- **Intake and elavation are seperate from one another**



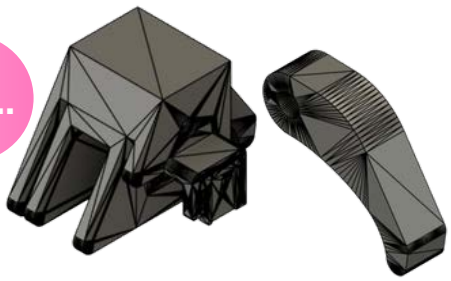
Version 3

Unique Aspects
- Spring hooks with Level 3 Ascent
- Linear actuator to tilt robot
- Extending roller intake
- Intake/outtake handoff
- Recycled-filament parts
- Compostable number plates

Manufacturing

3D Printers:

- Most of our custom parts are printed from PLA (Polylactic Acid), as its a lightweight and strong material that is also biodegradeable.
- We also used TPU (Thermoplastic Polyurethane) which allowed us to create flexible parts such as the intake rollers from our second iteration.





Laser Cutter:

- We used a laser cutter to cut our sideplates out of sheets of BC-grade plywood, as it was very quick and precise while still being strong enough.

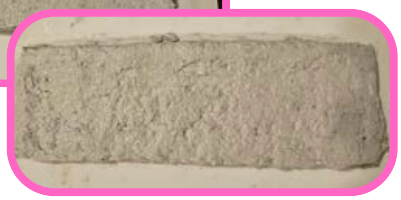




Number Plates:

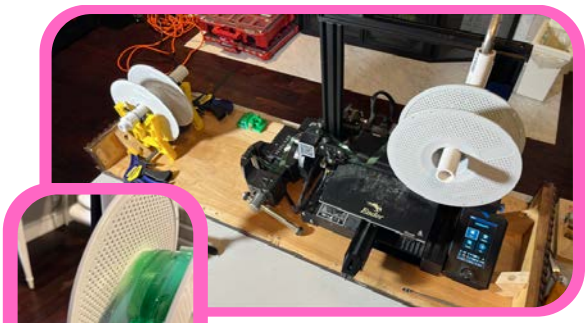
- Through a new process of using shredded paper scraps and milk-based glue, we were able to create our compostable and recyclable number plates!
- The glue we used is Casein glue, which can be made from curdling milk and putting it into an alkaline solution to have it stick to paper.





Recycled Filament:

- We converted an old unused 3D printer into a PETG filament recycler
- Used old extruder from printer to melt and extrude the repurposed plastic
- We were able to collect PETG-plastic liter bottles and convert the plastic into spools of filament that we used for prints.





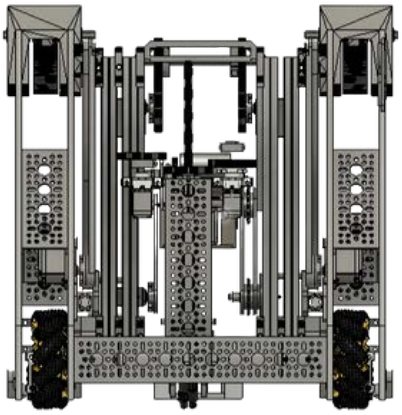
Robot Overview

Our Robot Design

- In the process of creating our robot, the use of CAD (computer-aided design) through Fusions 360 has been the backbone of our efforts, letting us design each mechanism, component, and system that is used on the robot.
- We were able to format our ideas into a testable model, and then use the model to manufacture.
- A prominent example of intensive design work would be our intake and outtake.

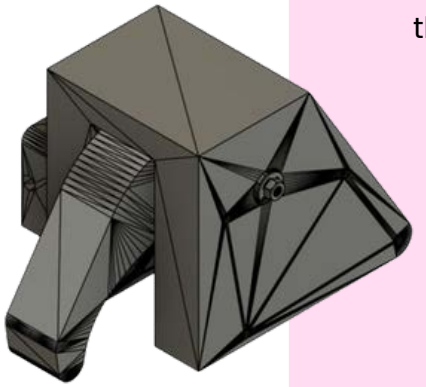
Drive Train:

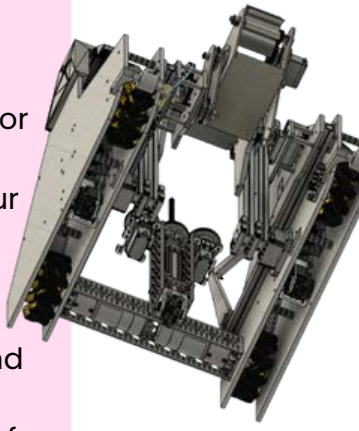
- 4 mecanum wheels
  - Motors are directly driving the wheels
- 435 rpm GoBilda Yellow Jacket motors
- Robot-centric drive instead of field-centric
  - We chose robot-centric based on preference
- Sideplates are wood instead of aluminum to make the robot lighter and faster to manufacture



Spring Hooks:

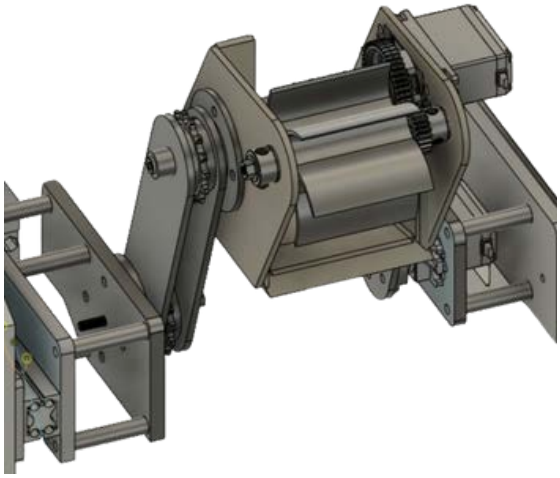
- Our ascent system is the most unique part of our robot.
  - They are fully 3D printed (besides the screws and springs) from PLA.
  - We make use of two sets of hooks and a linear actuator that work in conjunction to climb the submersible.
    - The linear actuator lifts half of the robot, putting our robot at the required angle for the linear slides to contact the first rung.
    - Our top hooks are mounted on the outtake linear slides, which we use to catch onto the first rung and lift our robot off the ground.
    - Our spring hooks are mounted on the side plates of our robot and are level with the top hooks when retracted. As the top hook pulls the spring hooks past the bars, the spring hooks are pushed into their casing and sprung back out, allowing the hook to slide past the bar and catch it.
- Once both sets of hooks catch onto the first rung, we repeat this process to achieve a Level 3 Ascent.





Intake:

- Our intake is capable of picking up samples and shifting them to our scoring mechanism, that will then place them into the buckets for scoring.
  - The movement for the pickup bucket is controlled by a virtual 4-bar, which is powered by a servo motor attached to the smaller, extended plates.
    - There are two sprockets that will be connected with chain that will aid in the movement of the intake.
  - The intaking mechanism consists of a 3D printed TPU roller that rolls to escort the samples into the bucket.
    - This is powered with by a servo motor with a speed gear ratio that allows us to pick up the samples quickly.
  - This mechanism is attached to side plate extensions on both sides that are connected to the actual side plates using standoffs. This makes mounting more secure.





Linear Slides:

- We have two sets of linear slides: the intake and the outtake.
  - For our slides, we decided to use models from MiSUMi, as they proved to be cost effective and lightweight.
  - The intake slides, which are MiSUMi SAR340's, are responsible for the pickup of samples from the floor or the submersible.
    - They are mounted horizontally and extend into samples for pickup, allowing for a faster cycle time.
  - The outtake slides, which are MiSUMi SAR330's, are responsible for scoring and ascent.
    - They are mounted vertically, with the ability to reach high basket and perform Level 3 Ascents.

Sideplates and Number Plates:

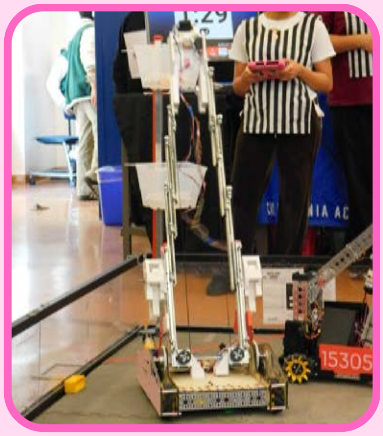
- Our side plates are unique compared to other teams because we use wood! We chose wood as it's lightweight, cheap, and much more eco-friendly than metal or polycarb.
- We used a laser cutter to manufacture them.
- They are custom designed using Fusion 360.
- For our number plates, we came up with a new method of making wood with similar properties to MDF using shredded up paper and milk-based casein glue.
- These new number plates are recyclable and compostable.





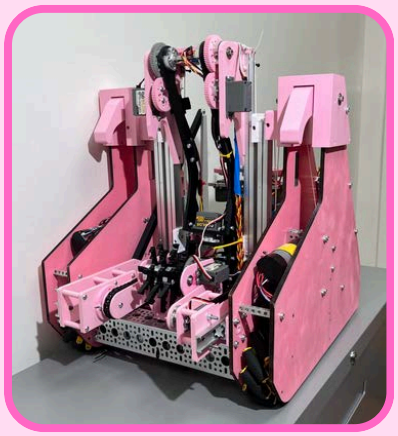
Both iterations of side plates!

Game Strategy



Autonomous Period (30 Seconds):

-Pick up the three yellow samples
-Score high baskets
- Park in observation zone
-Total: 27 Points



Teleoperated Period (2 Minutes):

- We focus on scoring samples in the high basket. Using our extended intake, we pick up a sample, either from around the field or in the submersible zone, drive over to the basket, extend our outtake linear slides, rotate our outtake, and drop in the sample. Alternatively, we can drop the sample to our human player, make specimens, and score those on the submersible.

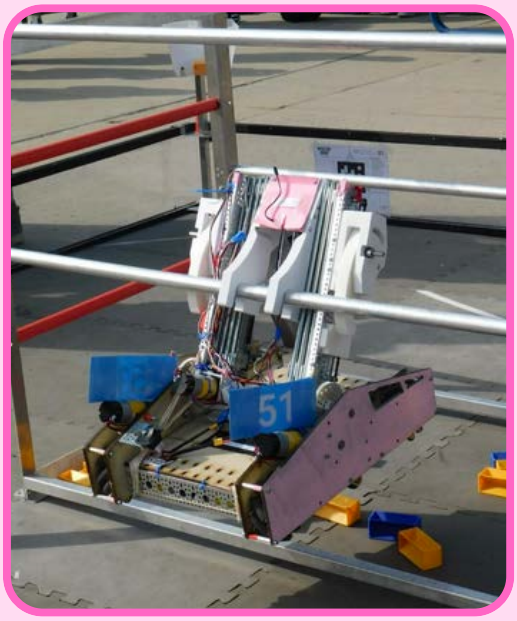
Last 30 seconds: End game

- Level 3 ascent (30 points)



Alliance Strategies

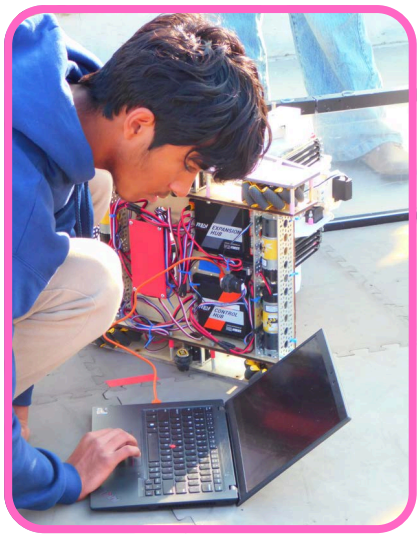
During our meets and our alliances, we changed our strategy and our way of working together with our allaince based on the strengths of our teams. For example, when we teamed up with Team 24777, we found that their scoring was more efficient, so we changed our strategy to gather samples for them, which allowed us to maximize the points we scored during teleop.



Programming

Tele-Operated

Currently, we use a field-oriented mecanum drive to control our robot during Tele-Op. We felt our drivers were more familiar with field-oriented drive, and it was for faster and more reliable while driving. This gave us more time to focus on drive practice and testing autonomous.





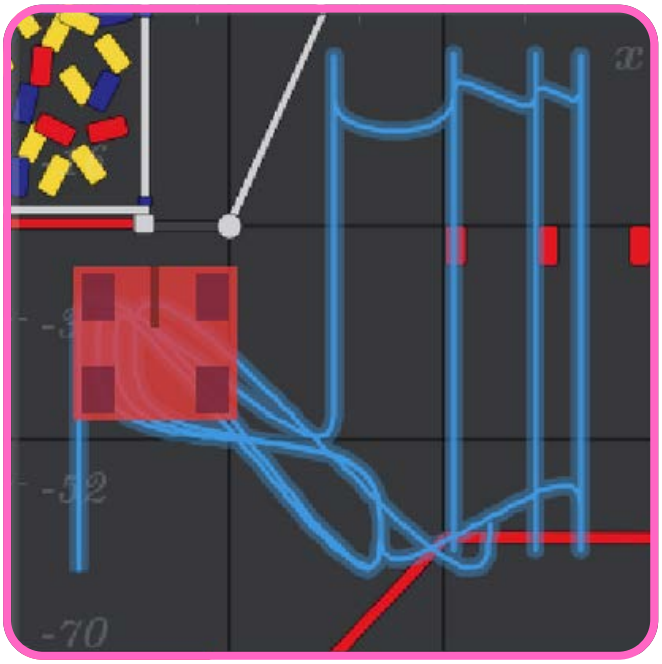
Our amazing programmers, Aadi and Maddie

Autonomous:

We use the **Roadrunner library** written by ACME Robotics to create complex spline paths for our robot to follow during the Autonomous period. We then follow Roadrunner paths using a **three dead wheel odometry setup**. This three wheel setup, while requiring more parts, is much more accurate than two wheel localization with the IMU. Instead of relying solely on the encoders built into motors, we also used odometry pods, which measure rotation.

Custom PID Loop:

We used **FTCLib**, developed by ARC Robotics, to create a PIDController that allowed us to tune PID coefficients via the FTC Dashboard by ACME Robotics. The use of the PID controller allows us to our vertical linear slides to accurately reach a desired position while ensuring even power distribution, preventing the motors from working against each other —an issue we encountered when using the built-in RUN_TO_POSITION mode from the FTC SDK.



Prototype Autonomous Path visualized using the MeepMeep Visualizer by ACME Robotics

Experiences & Learning

Events	What went well	What did not go well	Next Steps/What we learned
Meet 0	• Able to test how our team functions under pressure • Exposure to how "meets" operate	• Didn't have a fully functioning intake, ◦ Unable to score samples and specimens.	• Focus on developing an official drive team and drive practice. • Set better technical goals/deadlines. • Allocated more time for programming to troubleshoot
Meet 1	• Able to assemble an intake and outake • Scored low basket	• Intake and slides malfunctioned • Became pushbot • 3D printed parts failed	• Allocate more time to program with the robot
Meet 2	• Working intake and arm design	• Linear slides failed due to programming issues	• More time allocated to programming
Meet 3	• Working arm and intake • Able to score park autonomous and high basket samples • We were able to Level 2 hang consistently!	• Lack of driver practice and efficient design led to slow cycles • Lack of high scoring autonomous	• Increased Driver Practice • More Programming time • Better intake for ease of use • Make level 3 hang more consistent.

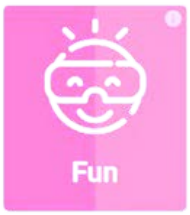














"Throughout my FTC experience on this team, I've learned so so much. Before I came to this team It was hard for me to have good **teamwork**, as well as come up with and execute my own ideas without lots of assistance. However with the help of my team and mentors, I've designed and **innovated** so many parts of our robot with our team members and I'm so proud of how far I've come!" -Michelle



"Over the course of this season, I have deepened my knowledge and **discovered** a great amount of robotics a great amount. I never felt left out in this team, and always felt a sense of **inclusion** and importatance with my work." - Suyash



"After attending many outreach events, especially the online lessons and our Elementary school outreach, it was such an honor to see the **impact** we left on the other kids! When we drove the robot, they all looked so happy and excited. I also had a ton of **fun** with all the members on this team. I've made so many new friends and learned a ton of different things. Even during stressful situations, we still had fun together as a team, and that's pretty cool!" - Ashley